Chih-Hai (Terry) Su

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Education

National Yang Ming Chiao Tung University (NYCU) — Hsinchu, Taiwan
*Ph.D. in Computer Science (Recipient of the **NSTC Doctoral Fellowship**)* — *Feb 2025 – Present (Expected 12/28)*

National Yang Ming Chiao Tung University (NYCU) — Hsinchu, Taiwan
B.S. in Computer Science (GPA: 3.95 / 4.30) — *Sep 2021 – Dec 2024*

Selected Publications

BoostMVSNeRFs: Boosting MVS-based NeRFs to Generalizable View Synthesis in Large-scale Scenes
Chih-Hai Su*, C.Y. Hu*, S.R. Tsai*, J.Y. Lee*, C.Y. Lin, Yu-Lun Liu. **SIGGRAPH 2024**.

See, Point, Fly: A Learning-Free VLM Framework for Universal Unmanned Aerial Navigation
C.Y. Hu*, Y.S. Lin*, Y. Lee, **Chih-Hai Su**, C.Y. Lin, K.W. Chen, S.R. Tsai, J.Y. Lee, T.W. Ke, Y.L. Liu. **CoRL 2025**.

Cepheid: Decoupling Scale from Structure for Cross-Family Motion Completion
Chih-Hai Su, Tsung-Wei Ke, Yu-Lun Liu. *Under review, ICLR 2027.*

TwinReason: Cross-Engine MLLM-Guided Inverse Physics from Monocular Video
Chih-Hai Su, F.E. Wang, N. Qiao, Y. Sun, et al. *Under review, WACV 2027.*
(* indicates equal contribution)

Research & Industry Experience

Amazon Development Center Taiwan Limited — Hsinchu, Taiwan
Applied Scientist Intern (CoRo Team) — *Feb 2025 – Aug 2025*

- **Inverse Physics**: Developed TwinReason to infer continuum dynamics from monocular RGB across Spring-Mass and MPM (DiffTaichi) simulators.
- **Efficient Search**: Reduced simulation queries from 220 to 40 (5.5×) vs. CMA-ES while jointly identifying categorical material regimes (jelly, plasticine, foam) and continuous physical parameters.
- **3D Reconstruction**: Built an uncalibrated tracking pipeline combining Grounded SAM 2, SAM3D, MoGE, and ARAP for dynamic 3D trajectory reconstruction.

NYCU Computational Photography Lab (Advisor: Prof. Yu-Lun Liu) — Hsinchu, Taiwan
Student Researcher — *Feb 2023 – Present*

- **Neural Rendering (SIGGRAPH '24)**: Proposed BoostMVSNeRFs, a training-free feed-forward framework addressing restricted viewport coverage and artifacts from limited input views via 3D visibility-guided greedy selection and multi-cost-volume rendering.
- **Dynamic Motion Prior**: Developed CEPHEID to decouple physical scale from motion structure using harmonic extension and dimensionless flow matching, enabling zero-shot sparse-to-dense mesh motion completion.

Phison Electronics Corporation — Miaoli, Taiwan
Machine Learning Intern — *May 2024 – Jan 2025*

- **Memory-Constrained Inference**: Ran large multimodal LLMs under tight GPU memory via DeepSpeed CPU offloading.
- **Inference Benchmarking**: Benchmarked vLLM, TensorRT, and tensor/pipeline parallelism (throughput, latency, memory).

Honors & Awards

Competitive Awards: ICPC Asia Regional Bronze Medal (2023), NYCU CS Annual Project Competition 1st Place (2024).

Fellowships & Grants: NSTC Doctoral Fellowship (2025–Present), NSTC Undergraduate Research Grant (2024).

Technical Skills

Languages: C/C++, Python, Bash, CUDA
Systems & Inference: vLLM, TensorRT, DeepSpeed, multi-GPU parallelism, ZeroMQ
ML & Vision: PyTorch, NeRF, 3DGS, Flow Matching, 3D Reconstruction, VLMs